

Project Strategy: RAG + Marketing Budget Recommender

Synthetic business documentation library for a campaign recommendation demo.

1. Project Strategy

This project demonstrates a hybrid marketing intelligence system: a quantitative recommendation engine estimates budget allocation and channel priority, while a RAG layer explains the recommendation in business language using curated PDF knowledge plus the campaign CSV.

Recommended system flow

- Ingest historical campaign data from CSV and normalize fields such as acquisition cost, channel, audience, campaign goal, segment and date.
- Create performance features such as ROI, conversion rate, engagement, click-through proxy, campaign duration and cost tiers.
- Generate recommendations by segment, audience and campaign objective: prioritized channels, budget range, expected ROI and rationale.
- Index this document library as PDF knowledge for RAG so the assistant can explain the rules, assumptions, benchmarks and caveats.
- Expose an analyst-facing chatbot that can answer questions such as: which channel should I prioritize for Technology Product Launch campaigns?

Reference metrics from the CSV

Channel	Campaigns	Avg ROI	Conv. Rate	Engagement	Avg Cost
Instagram	75,101	4.01	8.0%	5.51	\$7,726
Twitter	74,653	4.00	8.0%	5.50	\$7,774
Facebook	75,164	3.99	8.0%	5.48	\$7,745
Pinterest	75,082	0.72	8.0%	1	\$7,770

Segment	Campaigns	Avg ROI	Conv. Rate	Engagement	Avg Cost
Technology	59,840	3.19	8.0%	4.37	\$7,763
Health	60,086	3.19	8.0%	4.38	\$7,738
Fashion	59,940	3.18	8.0%	4.38	\$7,763
Home	60,077	3.17	8.0%	4.35	\$7,761
Food	60,057	3.16	8.0%	4.37	\$7,744

Recommendation logic for the MVP

The MVP should combine two simple layers: a rules-based scoring model and a budget sizing heuristic. This keeps the project explainable and easy to demo before introducing more advanced models such as contextual bandits or constrained optimization.

Component	MVP behavior	RAG explanation source
Channel score	Weighted score = 45% ROI + 25% conversion + 20% engagement + 10% volume stability	Channel Performance Benchmark Report

Component	MVP behavior	RAG explanation source
Budget tier	Map recommended spend to Low, Medium or High based on historical acquisition cost and expected ROI lift	Budget Allocation Framework
Business rationale	Explain why a channel was selected and what risk should be monitored	Marketing Strategy Playbook and Campaign Guidelines
Fallback	Use segment-level benchmarks when audience-goal combinations have low support	Customer Segment Insights

Target deliverable

The final demo can be positioned as Campaign Intelligence Agent: an AI assistant that recommends how to distribute paid media budget across Instagram, Facebook, Twitter and Pinterest by segment, audience and objective, and explains each recommendation with grounded business documents.